Supervised Deep Learning Under Label Noise

Reproduction of LabelWave & Entropy-Aware Loss Extension (EA-CPL)

ICLR 2024 Reproduction | **Novel EA-CPL Loss** | **CIFAR-10 · 40% Noise**

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The Problem: Label Noise in Deep Learning

Networks Memorize Noise



Zhang et al. proved DNNs perfectly memorize random labels if trained long enough — standard cross-entropy provides no defense.

Real Noise is Structured



Crowdsourcing errors cluster on hard, ambiguous samples — the exact cases where correct labels matter most.

Benchmarks Are Affected



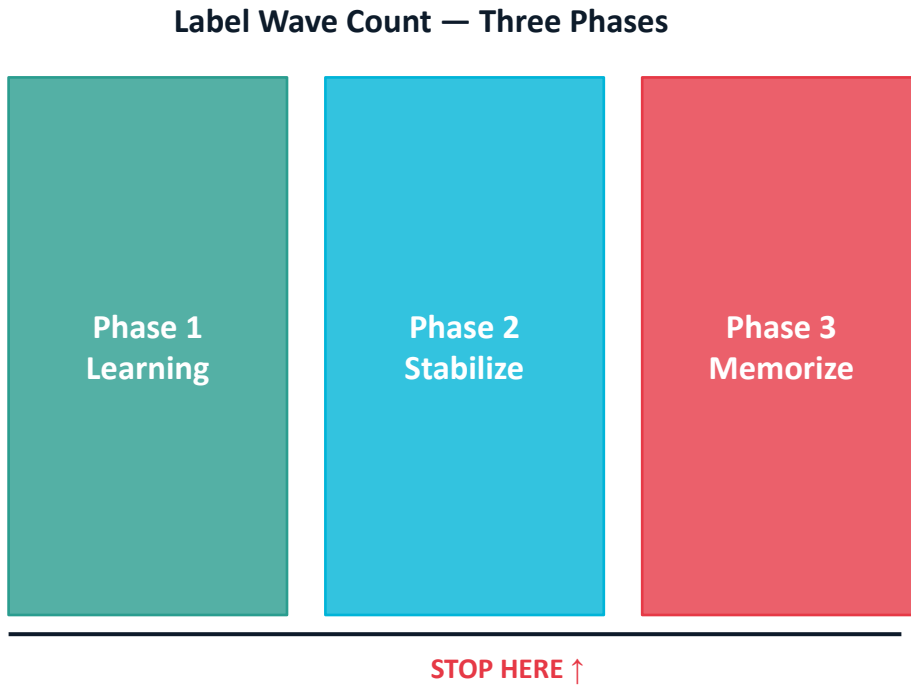
Northcutt et al. found pervasive label errors in popular test sets, distorting which methods appear best.

Training phases: Early Learning → Stabilization → Memorization (noise fitting)

LabelWave: Early Stopping Without Validation Data (ICLR 2024)

Core Idea

- Track how often each sample's predicted class changes between consecutive epochs
- Label Wave Count Wt = # samples that flip predictions epoch-to-epoch
- Spike ⟶ Decrease ⟶ Rising again signals start of memorization phase
- Stop training when count crosses adaptive threshold (mean + k·std)
- No clean validation set, no extra model, no noise-rate knowledge needed



Key advantage: detects memorization onset through temporal prediction instability

Our Extension: Entropy-Aware Confidence-Penalized Loss (EA-CPL)

Motivation

LabelWave stops training early, sacrificing clean-label learning.

EA-CPL Goal: Change what the model learns throughout training by penalizing overconfident predictions — without stopping.

Key insight: High-confidence predictions on noisy examples drive memorization. Suppress those specific gradient contributions.

How It Works

1 **Compute entropy**
$H(p)$ from softmax output

2 **Normalize → κ**
$\kappa = 1 - H(p)/\log C \in [0,1]$

3 **Per-sample smoothing**
$\alpha_i = clip(\alpha_0 + \lambda \cdot \kappa_i, 0, 0.9)$

4 **Soft target**
$\tilde{q}_{i,c}$ *distributes* α_i *over classes*

5 **EA-CPL Loss**
$L = -(1/N) \sum \sum \tilde{q}_{i,c} \log p_{i,c}$

Parameters: $\alpha_0 = 0.1$ (base smoothing), $\lambda = 0.5$ (penalty weight) — smoothing range: 0.1–0.6

Experimental Results

83.42%
LabelWave Repro
(50 ep, CPU)

79.94%
CE Baseline
(25 ep, GPU T4)

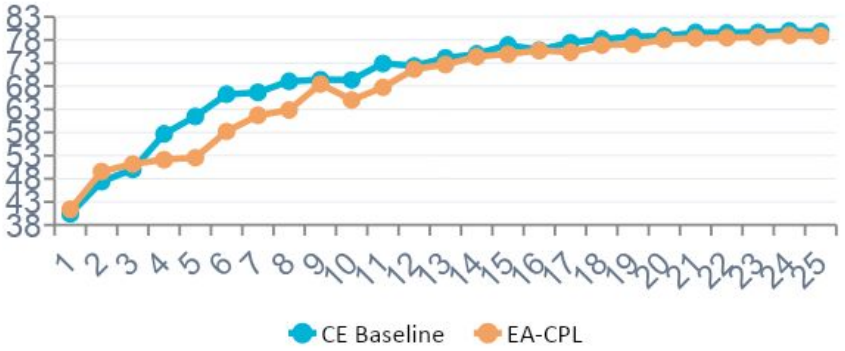
79.00%
EA-CPL Ours
(25 ep, GPU T4)

78.40%
Paper's LabelWave
(200 ep, GPU)

LabelWave Reproduction — Phase Checkpoints

Epoch	Wave Count	Test Acc.	Phase
2	30,233	52.47%	Learning Spike
27	9,247	82.21%	Stabilized
28	9,312	81.26%	Memorize Onset
32	9,158	83.42%	**Peak Acc.**

EA-CPL vs. CE Baseline — Test Accuracy



Analysis: Why EA-CPL Did Not Beat the Baseline

01 **Entropy ≠ Noise Indicator (Symmetric Noise)**

Under symmetric noise, both clean and memorized examples can produce high-confidence (low-entropy) predictions. EA-CPL penalizes both equally — slowing clean learning as much as noise fitting. The signal is blind to label quality.

02 **Convergence Penalty in Mid-Training**

The gap was largest at epochs 4–15 (CE: ~67%, EA-CPL: ~59%). Smoothed targets reduce gradient sharpness uniformly, causing EA-CPL to simply learn slower — not more robustly.

03 **Untuned Hyperparameters**

$\alpha_0 = 0.1$ and $\lambda = 0.5$ were principled but never empirically optimized. A grid search could substantially change the outcome.

04 **Entropy vs. Loss-Based Signals**

Unlike Co-teaching's small-loss heuristic, entropy ignores the label entirely. Small loss measures agreement between prediction and label — a directly relevant noise signal.

A negative result is still informative: entropy alone is insufficient as a per-sample noise proxy under symmetric corruption.

Conclusions & Future Directions

What We Found

- LabelWave reproduced: all 3 phases observed. Peak 83.42% (vs. paper's 78.4%) — run ended before deep memorization.
- EA-CPL scores 79.00% vs CE baseline's 79.94% — a −0.94 pp gap.
- Cosine annealing over 25 epochs implicitly prevents overfitting, explaining why our CE baseline already edges out LabelWave.
- Prediction entropy under symmetric noise cannot distinguish memorized noise from confident clean predictions.

Next Steps

- Test EA-CPL under asymmetric / instance-dependent noise where entropy may better correlate with label quality.
- Combine EA-CPL + LabelWave: entropy smoothing may delay the wave count spike → later, better stopping point.
- Run full 200-epoch GPU reproduction with early stopping to directly validate the paper's 78.4% claim.
- Hyperparameter grid search over α_0 and λ — current values were principled but unoptimized.